CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bus goes from A to B at 30 km/h and returns along the same road at 45 km/h. The average speed for the whole journey (km/h) is: 1

(a) 36 **(b)** 38 **(c)** 35 **(d)** 37

2. The average monthly salary of 23 employees is ₹75 thousand. An employee earning ₹36 thousand leaves and a new employee joins, and the average rises by ₹3 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 102 **(b)** 110 **(c)** 108 **(d)** 105

3. The average of 51 numbers is 71. The average of all 78 numbers (including 27 more numbers) is 53. The average of the 27 additional numbers is: 1

(a) 21 **(b)** 19 **(c)** 18 **(d)** 20

4. A shop earns a profit of 32% on sales of ₹4000 and 12% on sales of ₹6000. The overall profit percentage on total sales is: 1

(a) 22% **(b)** 20% **(c)** 19% **(d)** 21%

5. 780 mL of a 70% sugar solution is mixed with 1440 mL of a 33% sugar solution. The concentration of the mixture is: 1

(a) 45% **(b)** 48% **(c)** 47% **(d)** 46%

6. The average marks of 80 students is 93 and of another 100 students is 75. The average of all 180 students is: 1

(a) 82 **(b)** 85 **(c)** 83 **(d)** 84

7. A group of 42 people has average weight 110 kg. Another group has average weight 72 kg. Together the average weight is 91 kg. The number of people in the second group is: 1

(a) 44 **(b)** 43 **(c)** 41 **(d)** 42

8. In what ratio must a 26% solution and a 64% solution be mixed to obtain a 49% solution? 1

(a) 15 : 24 **(b)** 16 : 23 **(c)** 15 : 23 **(d)** 23 : 15

9. **Assertion (A):** The combined average of two groups lies between the smaller and the larger group average. 1

Reason (R): If two groups are of the same size, the combined average is the simple mean of the two group averages.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Mixing 200 mL of a 10% solution with 300 mL of a 20% solution gives a 16% solution. 1
Reason (R): The concentration of a mixture is the simple mean of the two concentrations, $\frac{10+20}{2} = 15\%$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A cyclist rides from home to a market at 44 km/h and returns by the same route at 77 km/h. Find the average speed for the whole trip. 2

12. The average monthly salary of 23 employees is ₹64 thousand. An employee earning ₹38 thousand leaves and a new employee joins; the average rises by ₹3 thousand. Find the new employee's monthly salary. 2

13. A chemist has a 41% acid solution and a 72% acid solution. In what ratio must they be mixed to get a 64% solution? 2

14. A cyclist rides from home to a market at 30 km/h and returns by the same route at 90 km/h. Find the average speed for the whole trip. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a class, 70 students have an average of 67 marks and the remaining students have an average of 51 marks. The class average is 65. Find the number of remaining students. 3

16. A final score combines Maths (score 52, weight 6), Science (score 61, weight 8), English (score 72, weight 6), Hindi (score 82, weight 4). Find the weighted average score. 3

17. The mean of 14 numbers is 32. Two more numbers 51 and 29 are added. Find the new mean and say whether it rises or falls. 3

OR

In a class, 56 students have an average of 97 marks and the remaining students have an average of 87 marks. The class average is 95. Find the number of remaining students. 3

18. A final score combines Maths (score 55, weight 6), Science (score 61, weight 2), English (score 72, weight 3), Hindi (score 53, weight 6). Find the weighted average score. 3

OR

The mean of 11 numbers is 38. Two more numbers 52 and 37 are added. Find the new mean and say whether it rises or falls. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A juice vendor in Mumbai mixes 4 litres of orange squash containing 30% pure juice with 6 litres of squash containing 20% pure juice. 5

(i) Find the pure juice in the first squash.

(ii) Find the pure juice in the second squash.

(iii) Find the concentration of the mixture.

(iv) How much water must be added to the mixture to make it a 20% solution?

20. A trader mixes rice costing ₹120 per kg and rice costing ₹180 per kg in the ratio 4:2. Find the cost price per kg of the mixture and the selling price per kg to earn a profit of 20%. 5

OR

A family spent ₹4600 in January and ₹7500 in February. Food took 38% of the January total and 24% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

END OF QUESTION PAPER — Total: 40 Marks